PROJECT PROGRESS REPORT

PROJECT PROGRESS REPORT
(Reporting No. 05/10)
***Reporting Period: From** 13/08/2026 **To** 20/08/2026*

1. Student Information

Student Name	PATEL MAN VISHNUBHAI.
Enrollment Number	25IC03CA053
Program/Branch	MCA (CA)/ICA
Semester	3
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2. Project Details

Type	Research Project
Department/Domain	MCA CA(DS/ML) / AI & ML
Mentor Name	Srikant Singh
Start Date	08-06-2026
End Date	03-10-2026

3. Project Title

- OpenCV - Based System for Crack Identification and Width Estimation

4. Progress During the Reporting Period

A. Previous Reporting Review

Planned Task	Status	Remarks
Install & Import Libraries	Completed	Required Python and OpenCV libraries were imported successfully for image processing and crack analysis.
Project Configuration	Completed	Project name, output configuration and required processing settings were configured.
Output & Mask Directories	Completed	Directories for annotated images, crack masks and processed images were created successfully.
OpenCV Parameters	Completed	Gaussian blur, black-hat transformation, morphological kernel and minimum crack-area parameters were configured.
pixel-to-mm calibration	In Progress	Pixel-based measurement is implemented. Physical mm calibration requires a known reference measurement.
RISK CLASSIFICATION THRESHOLDS	In Progress	Risk classification thresholds are being configured based on estimated crack width and analysis results.
UPLOAD MULTIPLE IMAGES	Pending	Multiple input images can be uploaded for batch crack analysis.
CRACK LENGTH Management	Pending	Crack length extraction and measurement are being tested and refined for different crack shapes.
IMAGE ANALYSIS	Pending	Image preprocessing, crack detection, segmentation and measurement are being tested on uploaded images.

B. Work Completed During This Reporting Period

1. **Activities/tasks completed**

- Completed project setup, required library installation, image-upload functionality, output-directory configuration, and initial crack-image pre-processing.

2. **Modules/features developed**

- Developed modules for image preprocessing, crack detection, crack-mask generation, multiple-image upload, crack length management, and result storage.

3. **Research or literature work completed**

- Reviewed existing computer-vision techniques for crack detection, including Gaussian filtering, black-hat transformation, thresholding, morphological operations, crack length measurement, and pixel-to-mm calibration.

4. **Design/development/testing activities**

- Designed the OpenCV processing pipeline and configured Gaussian blur, black-hat kernel, morphological operations, minimum crack-area filtering, and calibration parameters. Tested the system using multiple crack images.

5. **Results or outputs achieved**

- Successfully generated processed images, crack masks, and annotated results. Initial crack detection and segmentation were achieved, while crack-width calibration and risk classification are currently being refined and validated.

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C. Evidence / Deliverables Produced

Sr.no	Deliverable / Evidence	Description
1	Google Colab Source Code	Complete OpenCV crack detection and width estimation code.
2	Input Images	Multiple crack images used for testing.
3	Processed Images	Preprocessed and enhanced crack images.
4	Crack Masks	Binary masks showing detected crack regions.
5	Annotated Images	Images showing detected cracks and analysis results
6	Crack Measurement	Crack length and width measurement outputs.
7	Calibration	Pixel-to-mm calibration configuration and testing.
8	Risk Classification	Crack severity/risk classification results.

D. Current Project Progress

Overall Project Completion: 30%

Sr. No.	Project Component	Status	Completion
1	Project Setup & Configuration	Completed	**100%**
2	Library Installation & Import	Completed	**100%**
3	Image Upload	Completed	**100%**
4	Image Preprocessing	In Progress	**70%**
5	Crack Segmentation	In Progress	**10%**
6	Crack Detection	Pending	**0%**
7	Crack Width Estimation	Pending	**0%**
8	Risk Classification	Pending	**0%**
9	Result Visualization	Pending	**0%**
10	CSV Report & ZIP Download	Pending	**0%**

E. Challenges / Issues Faced

- **Image Quality Variation:** Different crack images have variations in lighting, shadows, surface texture, resolution, and camera distance, which can affect crack detection accuracy.
- **Crack Segmentation:** Separating actual cracks from concrete texture, noise, shadows, and other dark regions is challenging.
- **Parameter Selection:** Selecting suitable Gaussian blur, black-hat kernel, morphological kernel, threshold, and minimum crack-area values requires testing and tuning.
- **Crack Width Estimation:** Accurate crack-width measurement requires reliable segmentation and skeletonization of the detected crack.
- **Pixel-to-MM Calibration:** Converting pixel measurements into real-world millimeters requires a proper reference object and consistent camera conditions.
- **Risk Classification:** Defining appropriate Low, Medium, and High-risk thresholds requires reliable crack-width measurements and validation.
- **Multiple Image Processing:** Processing multiple images consistently requires proper file handling, output management, and error handling.

F. Actions Taken / Solution Implemented

- Installed and configured all required Python and OpenCV libraries.
- Configured project directories for processed images, crack masks, and annotated results.
- Applied Gaussian blur to reduce image noise and surface variations.
- Configured black-hat transformation to enhance dark crack regions.
- Applied morphological operations to improve crack segmentation and remove noise.
- Set minimum crack-area filtering to eliminate unwanted small regions.

- Configured pixel-to-mm calibration for physical crack-width measurement.
- Prepared the system configuration for crack detection, width estimation, and risk classification.
- Planned multiple-image processing and result storage for further testing and validation.

G. Plan for Next Reporting

Sr. No.	Planned Activity	Expected Deliverable	Target Date
1	Image upload, preprocessing and module completion	Working image upload and pre-processing module	27/August/2026
2	Crack segmentation, detection and width estimation	Crack masks, detected cracks, length and width measurements	03/September/2026
3	Risk classification, visualization and reporting	Risk classification, annotated results and CSV report	10/September/2026
4	Final testing, optimization and documentation	Tested and optimized system with complete documentation	17/September/2026
5	Final demonstration, presentation and viva preparation	Final project, presentation, report and viva-ready system	24/September/2026

H. Learning / Skills Acquired

- Learned Python and OpenCV libraries for computer-vision-based image processing.
- Gained knowledge of image preprocessing techniques such as grayscale conversion, Gaussian filtering, and noise reduction.
- Learned black-hat morphological transformation for enhancing dark crack regions.
- Developed understanding of thresholding, morphological operations, and crack segmentation.

- Learned techniques for crack detection, skeletonization, and crack length estimation.
- Gained knowledge of pixel-based crack width measurement and pixel-to-mm calibration.
- Learned basic risk classification based on crack measurement results.
- Learned result visualization and management of processed images and crack masks.

4. Tasks Accomplished

- Installed and configured the required Python and OpenCV libraries.
- Imported and verified all required project libraries.
- Configured the project structure and output directories.
- Configured OpenCV parameters for image preprocessing and crack enhancement.
- Implemented output folders for annotated images, crack masks, and processed images.
- Configured pixel-to-mm calibration parameters for crack-width measurement.
- Prepared the project framework for image upload and crack analysis.
- Prepared the system for crack segmentation, detection, width estimation, risk classification, and result reporting.

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5. Tasks in Progress / Next Plan

- Develop multiple-image upload and image preprocessing functionality.
- Implement crack segmentation and accurate crack detection.
- Develop crack length and width estimation using skeletonization and calibration.
- Implement risk classification based on crack measurements.
- Develop result visualization with annotated images and crack masks.
- Implement CSV report generation and ZIP download functionality.
- Perform testing, error handling, parameter optimization, and accuracy validation.

6. Learnings / Skills Acquired

- Learned Python, NumPy, Pandas, OpenCV, and Scikit-image for image processing and analysis.
- Gained practical knowledge of image preprocessing, Gaussian filtering, and noise reduction.
- Learned black-hat transformation and morphological operations for crack enhancement and segmentation.
- Developed understanding of crack detection, skeletonization, and crack length estimation techniques.
- Learned pixel-based crack-width measurement and pixel-to-mm calibration concepts.
- Gained experience in configuring image-analysis parameters and managing processing outputs.

References:

- **Deng, J., Singh, A., Zhou, Y., Lu, Y., & Lee, V. C.-S. (2022).** *Review on computer vision-based crack detection and quantification methodologies for civil structures.* Construction and Building Materials, 356, 129238.
- **Yu, L. (2022).** *Intelligent Crack Detection and Quantification in the Concrete Bridge: A Deep Learning-Assisted Image Processing Approach.* Advances in Civil Engineering, 2022, 1813821.
- **Dung, C. V. (2019).** *Autonomous concrete crack detection using deep fully convolutional neural network.* Automation in Construction.
- **Mohan, A., & Poobal, S. (2018).** *Crack detection using image processing: A critical review and analysis.* Alexandria Engineering Journal, 57(2), 787–798.

Evaluation (To be filled by Mentor)

Criteria	Bloom's Level	Excellent (20)	Good (15)	Average (10)	Poor (5)	Marks
Progress as per Previous Plan	L3– Apply	All planned tasks completed	Most completed	Partial completion	Very little progress	
Technical / Research Implementation	L4– Analyze	Excellent implementation with clear evidence	Good implementation	Basic implementation	Insufficient evidence	
Quality of Deliverables	L5– Evaluate	High-quality, validated deliverables	Good quality	Acceptable	Poor / incomplete	
Problem Solving & Challenges	L4– Analyze	Challenges clearly identified and effectively solved	Most issues addressed	Limited solutions	No clear effort	
Planning for Next Phase	L6– Create	Clear, realistic and measurable plan	Mostly clear plan	Basic plan	No proper plan	
Documentation & Presentation	L3– Apply	Well-structured and professional	Minor issues	Average clarity	Poor documentation	
Total marks out of 100						

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Evaluation

Student Signature	Mentor (Institute) Signature

Mentor (Institute) Remarks

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